

HW 1

1) A)
(i) system = atom + photon

Objective: we need to trace out the photon
(partial trace: $\text{tr}_E(A \otimes B) = A \text{tr}(B)$)
and prove that the evolution of the atom
alone can be represented by Kraus operators.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{--- atom}$$

$$|\phi\rangle = |\psi\rangle|0\rangle = (\alpha|0\rangle + \beta|1\rangle)|0\rangle \quad \text{--- atom entangled with photon}$$

Now, $|\phi\rangle \xrightarrow{U} |\phi'\rangle$ where

$$|\phi'\rangle = \alpha|00\rangle + \beta[\sqrt{1-P}|10\rangle + \sqrt{P}|01\rangle]$$

(spontaneous emission
happens with a
probability P)

Simplifying the state:

$$\Rightarrow \alpha|00\rangle + \beta\sqrt{1-P}|10\rangle + \beta\sqrt{P}|01\rangle$$

$$|\phi'\rangle = \underbrace{(\alpha|0\rangle + \beta\sqrt{1-p}|1\rangle)}_{\text{atom}} \underbrace{|0\rangle}_{\text{photon}} + \underbrace{\beta\sqrt{p}|0\rangle}_{\text{atom}} \underbrace{|r\rangle}_{\text{photon}}$$

Now, to trace out the photon we need to compute the density matrix of the new state

$$= |\phi'\rangle\langle\phi'|$$

$$|\phi'\rangle = \underbrace{(\alpha|0\rangle + \beta\sqrt{1-p}|1\rangle)}_{|\psi_1'\rangle|0\rangle} + \underbrace{\beta\sqrt{1-p}|0\rangle}_{|\psi_2'\rangle|r\rangle}$$

$$|\phi'\rangle\langle\phi'|$$

$$= (|\psi_1'\rangle|0\rangle + |\psi_2'\rangle|r\rangle)(|\psi_1'\rangle|0\rangle + |\psi_2'\rangle|r\rangle)^\dagger$$

$$= (|\psi_1'\rangle|0\rangle + |\psi_2'\rangle|r\rangle)(\langle\psi_1'|\langle 0| + \langle\psi_2'|\langle r|)$$

$$= |\psi_1'\rangle|0\rangle\langle\psi_1'|\langle 0| + |\psi_1'\rangle|0\rangle\langle\psi_2'|\langle r|$$

$$+ |\psi_2'\rangle|r\rangle\langle\psi_1'|\langle 0| + |\psi_2'\rangle|r\rangle\langle\psi_2'|\langle r|$$

$$(|a\rangle \otimes |b\rangle) \otimes (\langle c| \otimes \langle d|) = (|a\rangle \otimes \langle c|) \otimes (|b\rangle \otimes \langle d|)$$

$$\Rightarrow |\psi_1'\rangle \langle \psi_1'| |0\rangle \langle 0| + |\psi_1'\rangle \langle \psi_2'| |0\rangle \langle r|$$

$$+ |\psi_2'\rangle \langle \psi_1'| |r\rangle \langle 0| + |\psi_2'\rangle \langle \psi_2'| |r\rangle \langle r|$$

$$+r_{\text{photon}} \left(\begin{array}{cc} |\psi_1'\rangle \langle \psi_1'| |0\rangle \langle 0| & + |\psi_1'\rangle \langle \psi_2'| |0\rangle \langle r| \\ + |\psi_2'\rangle \langle \psi_1'| |r\rangle \langle 0| & + |\psi_2'\rangle \langle \psi_2'| |r\rangle \langle r| \end{array} \right)$$

$$= |\psi_1'\rangle \langle \psi_1'| +r(|0\rangle \langle 0|) + |\psi_1'\rangle \langle \psi_2'| +r(|0\rangle \langle r|) \\ + |\psi_2'\rangle \langle \psi_1'| +r(|r\rangle \langle 0|) + |\psi_2'\rangle \langle \psi_2'| +r(|r\rangle \langle r|)$$

$$= |\psi_1'\rangle \langle \psi_1'| \underbrace{\langle 0|0\rangle}_1 + |\psi_1'\rangle \langle \psi_2'| \underbrace{\langle r|0\rangle}_0 \\ + |\psi_2'\rangle \langle \psi_1'| \underbrace{\langle 0|r\rangle}_0 + |\psi_2'\rangle \langle \psi_2'| \underbrace{\langle r|r\rangle}_1$$

$$\rho_{\psi'} = |\psi_1'\rangle \langle \psi_1'| + |\psi_2'\rangle \langle \psi_2'|$$

Now, we know that

$$K_1 = |0\rangle \langle 0| + \sqrt{1-p} |1\rangle \langle 1| ; K_2 = \sqrt{p} |0\rangle \langle 1|$$

$$\text{So, } K_1 |\psi\rangle = (|0\rangle \langle 0| + \sqrt{1-p} |1\rangle \langle 1|) (\alpha |0\rangle + \beta |1\rangle) \\ = \alpha |0\rangle + \beta \sqrt{1-p} |1\rangle = |\psi_1'\rangle$$

similarly,

$$\begin{aligned} K_2 |\psi\rangle &= (\sqrt{p} |0\rangle \langle 1|) (\alpha |0\rangle + \beta |1\rangle) \\ &= \beta \sqrt{p} |0\rangle = |\psi'_2\rangle \end{aligned}$$

$$\therefore \rho_{\phi'} = |\psi'_1\rangle \langle \psi'_1| + |\psi'_2\rangle \langle \psi'_2|$$

$$\rho_{\phi'} = K_1 |\psi\rangle \langle \psi| K_1^\dagger + K_2 |\psi\rangle \langle \psi| K_2^\dagger$$

$$\therefore \mathcal{N}(\rho) = \rho_{\phi'} = K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger$$

... QED.

(ii) Objective: verify that $\mathcal{N}$ is a valid quantum channel.

If $\mathcal{N}$ were a valid quantum channel it must follow CPTP [Completely Positive; Trace Preserving]

$$= \text{tr} (K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger)$$

$$= \text{tr} (K_1 \rho K_1^\dagger) + \text{tr} (K_2 \rho K_2^\dagger)$$

$$= \text{tr} (\rho K_1^\dagger K_1) + \text{tr} (\rho K_2^\dagger K_2) \text{ trace is cyclic}$$

$$K_1 = K_1^\dagger = |0\rangle\langle 0| + \sqrt{1-p}|1\rangle\langle 1|$$

$$K_1^\dagger K_1 = |0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$$

Similarly,

$$K_2 = \sqrt{p}|0\rangle\langle 1| \text{ \& } K_2^\dagger = \sqrt{p}|1\rangle\langle 0|$$

$$K_2^\dagger K_2 = p \underbrace{|1\rangle\langle 0|0\rangle\langle 1|}_1 = p|1\rangle\langle 1|$$

Now,

$$\text{tr}(\rho K_1^\dagger K_1) + \text{tr}(\rho K_2^\dagger K_2)$$

$$= \text{tr}[\rho(|0\rangle\langle 0| + (1-p)|1\rangle\langle 1| + p|1\rangle\langle 1|)]$$

$$= \text{tr}[\rho \underbrace{(|0\rangle\langle 0| + |1\rangle\langle 1|)}_I]$$

$$= \text{tr}[\rho] \text{ which shows that the channel is trace preserving.}$$

$$\rho_{\phi'} = K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger$$

$$\begin{aligned} \langle \phi' | \rho_{\phi'} | \phi' \rangle &= \langle \phi' | K_1 \rho K_1^\dagger | \phi' \rangle \\ &\quad + \langle \phi' | K_2 \rho K_2^\dagger | \phi' \rangle \end{aligned}$$

$$\langle \phi' | K_1 \rho K_1^\dagger | \phi' \rangle + \langle \phi' | K_2 \rho K_2^\dagger | \phi' \rangle$$

$$|\varphi_1\rangle = K_1^\dagger |\phi'\rangle \quad \langle \varphi_1| = \langle \phi' | K_1$$

$$|\varphi_2\rangle = K_2^\dagger |\phi'\rangle \quad \langle \varphi_2| = \langle \phi' | K_2$$

So,

$$\langle \varphi_1 | \rho | \varphi_1 \rangle + \langle \varphi_2 | \rho | \varphi_2 \rangle$$

$$\therefore \rho \succeq 0 \quad [\text{density matrix}]$$

$$\langle \varphi_1 | \rho | \varphi_1 \rangle + \langle \varphi_2 | \rho | \varphi_2 \rangle \geq 0$$

Hence, it's completely positive.

For it to be linear

$$C(P_1 \rho_1 + P_2 \rho_2) = P_1 C(\rho_1) + P_2 C(\rho_2)$$

$$C(\rho) = K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger$$

$$\begin{aligned} C(P_1 \rho_1 + P_2 \rho_2) &= K_1 (P_1 \rho_1 + P_2 \rho_2) K_1^\dagger + \\ &\quad \underbrace{K_2 (P_1 \rho_1 + P_2 \rho_2) K_2^\dagger}_{\substack{K_2 P_1 \rho_1 K_2^\dagger + K_2 P_2 \rho_2 K_2^\dagger}} \\ &= \underbrace{K_1 P_1 \rho_1 K_1^\dagger + K_1 P_2 \rho_2 K_1^\dagger}_{\substack{K_1 P_1 \rho_1 K_1^\dagger + K_1 P_2 \rho_2 K_1^\dagger}} + K_2 P_1 \rho_1 K_2^\dagger + K_2 P_2 \rho_2 K_2^\dagger \end{aligned}$$

$$\begin{aligned}
 &= P_1 (K_1 P_1 K_1^+ + K_2 P_1 K_2^+) + P_2 (K_1 P_2 K_1^+ + K_2 P_2 K_2^+) \\
 &= P_1 C(P_1) + P_2 C(P_2)
 \end{aligned}$$

Hence, the channel is linear for probabilistic mixtures.

The channel $\mathcal{N}$ satisfies all the three conditions, therefore it is valid.

... QED.